

01 Vectors

1.1 What Is a Vector?

A vector has both magnitude and direction



Linear Algebra Made Easy and Visual | Open-source code & drafts: <https://github.com/Visualize-ML>

What You Will Learn

1

The definition and essence of vectors: magnitude + direction, distinct from scalars

2

Geometric intuition: reading magnitude and direction from an arrow

3

Row vectors vs. column vectors — arrangement, notation, and use cases

4

The transpose operation: turning a row vector into a column vector, and back

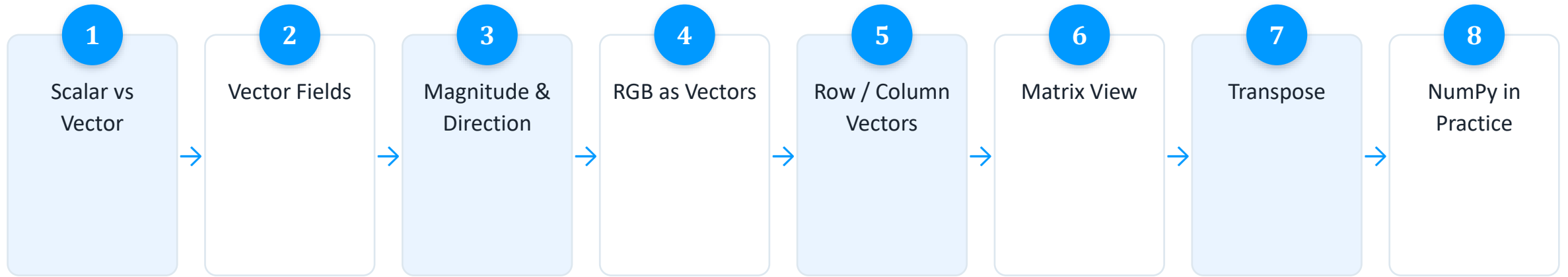
5

Creating and manipulating vectors with NumPy: indexing and slicing 1D & 2D arrays

6

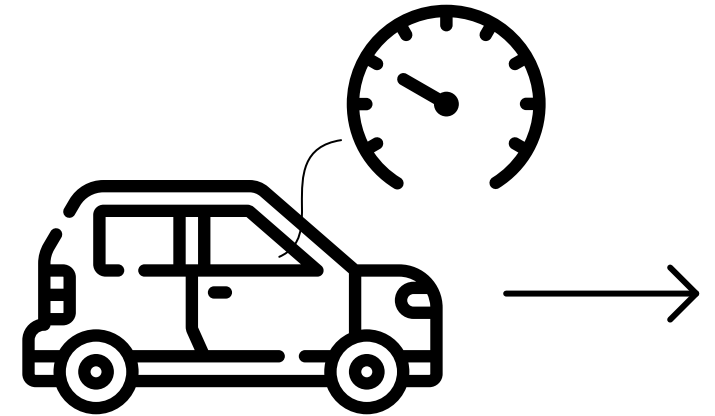
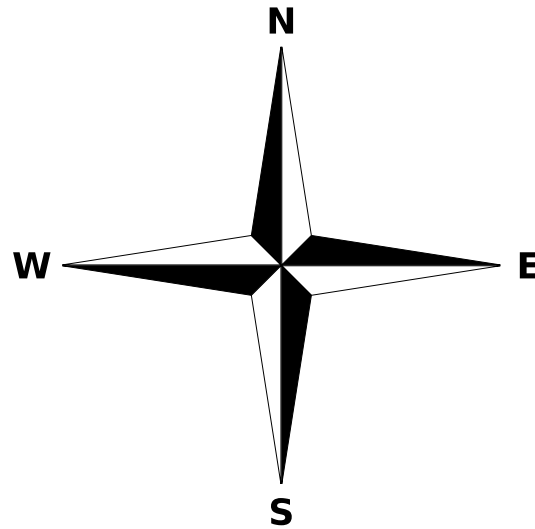
How vectors relate to matrices

How This Section Unfolds



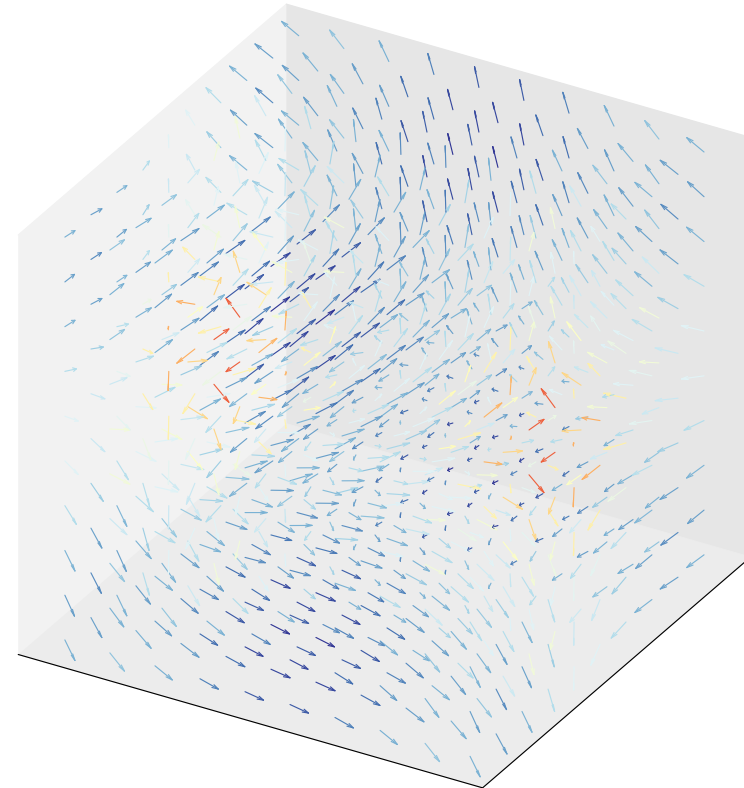
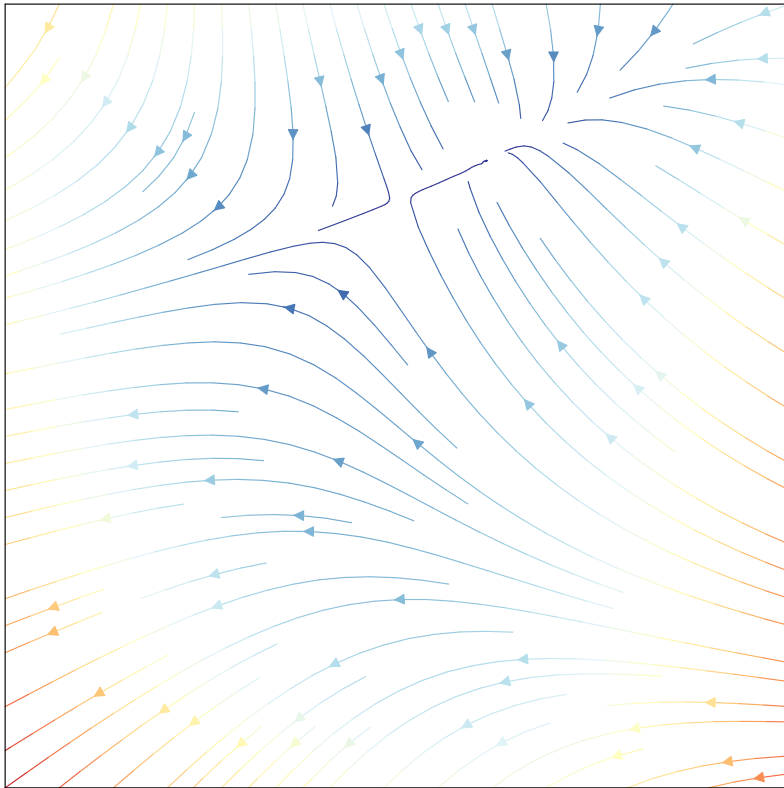
Scalar vs. Vector

- Scalar: magnitude only, no direction — e.g. a car's speed of 60 km/h.
- Vector: magnitude and direction together — e.g. “60 km/h due east.”
- Geometrically, a vector is a directed line segment.
- Same for displacement: “5 km” is a scalar; “5 km southwest” is a displacement (vector).



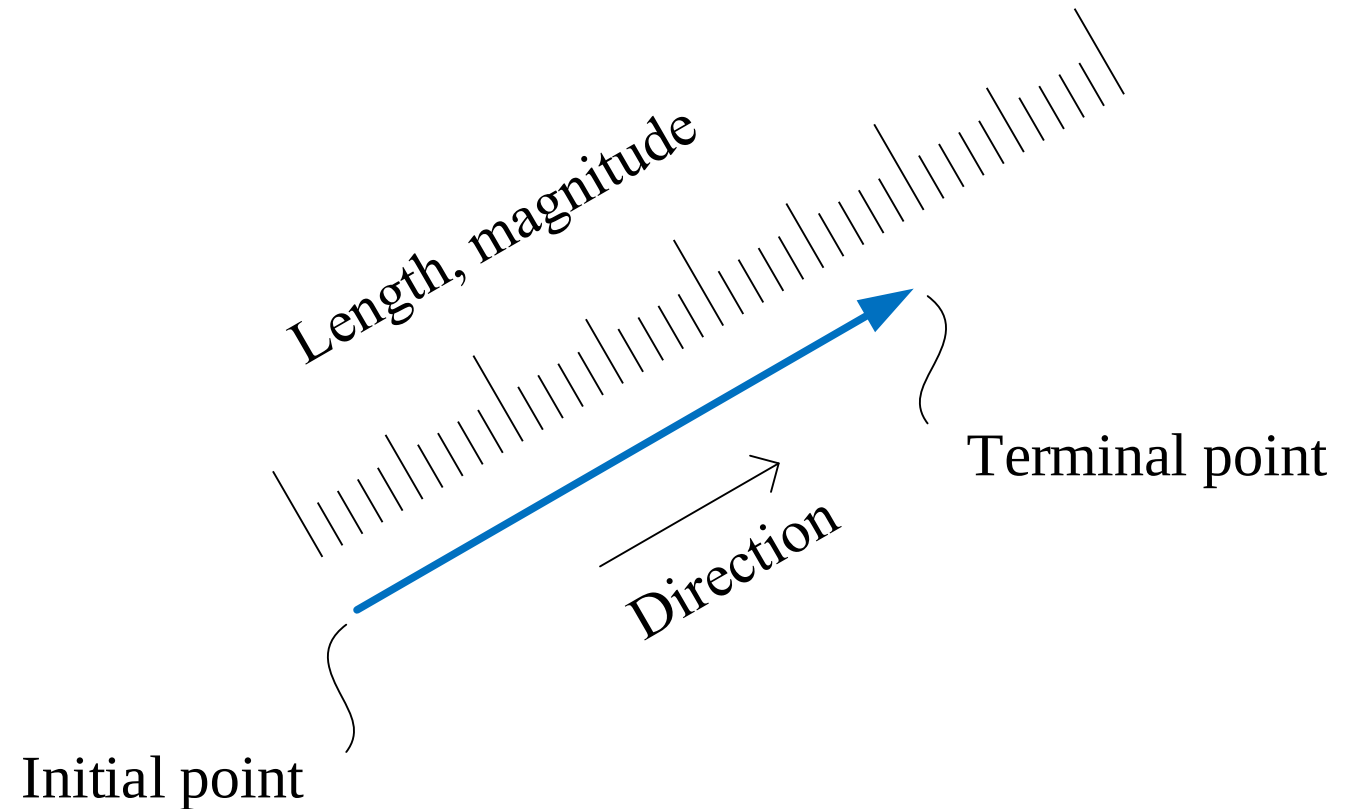
Vector Fields: Wind, Water, Gravity

- A vector field assigns a vector to every point in space, describing the magnitude and direction of some quantity there.
- Wind, water flow, gravity, and electromagnetic fields are all vector fields — a planar field (left) and a spatial field (right) are shown below.



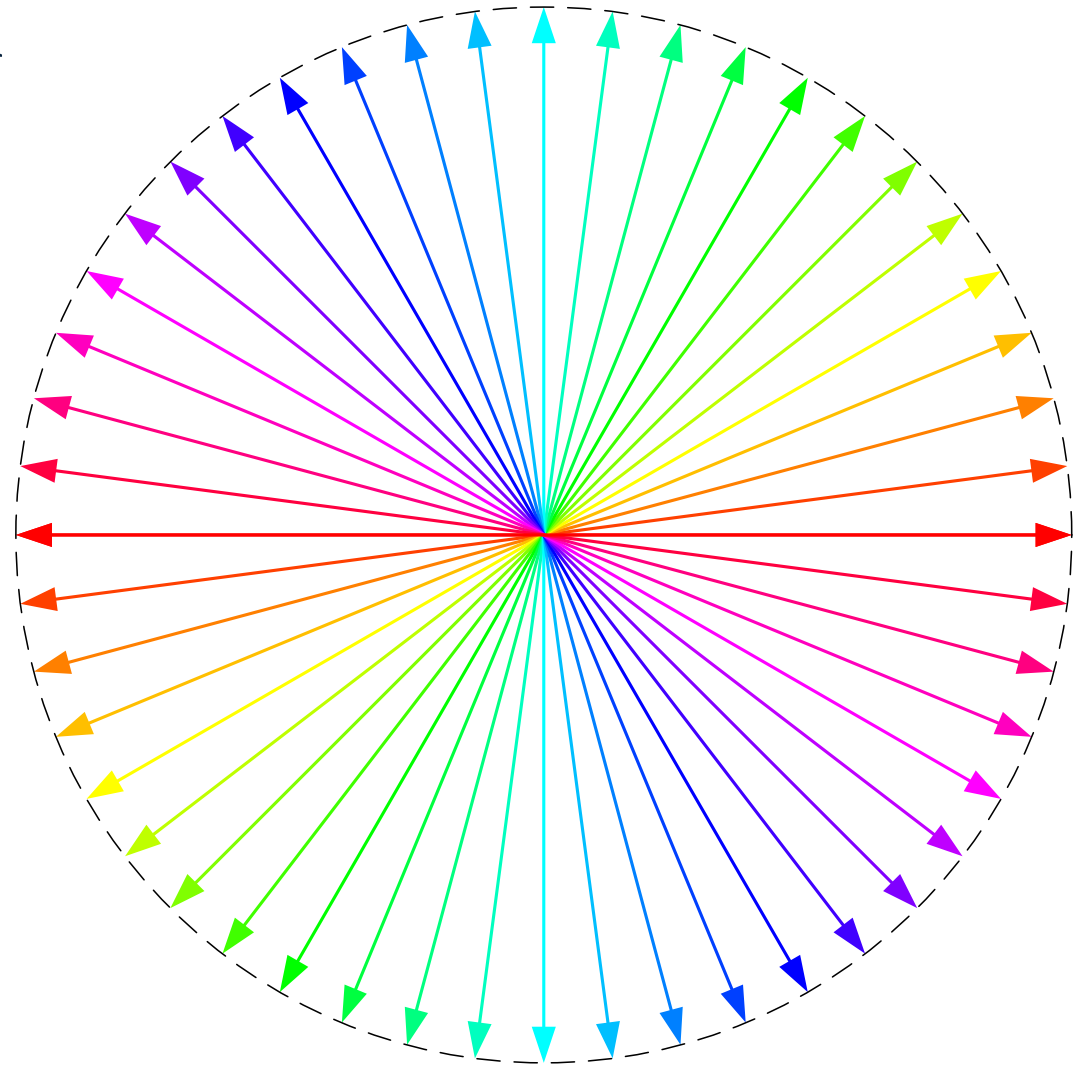
Initial Point, Terminal Point, Length

- Picture a vector as an arrow: its length is the magnitude, its heading is the direction.
- Magnitude (also called length): the straight-line distance from the initial point (tail) to the terminal point (head).
- The way the arrowhead points is the vector's direction.
- This arrow-like nature is why vectors are also called “vectors” in physics — literally, carriers of direction.



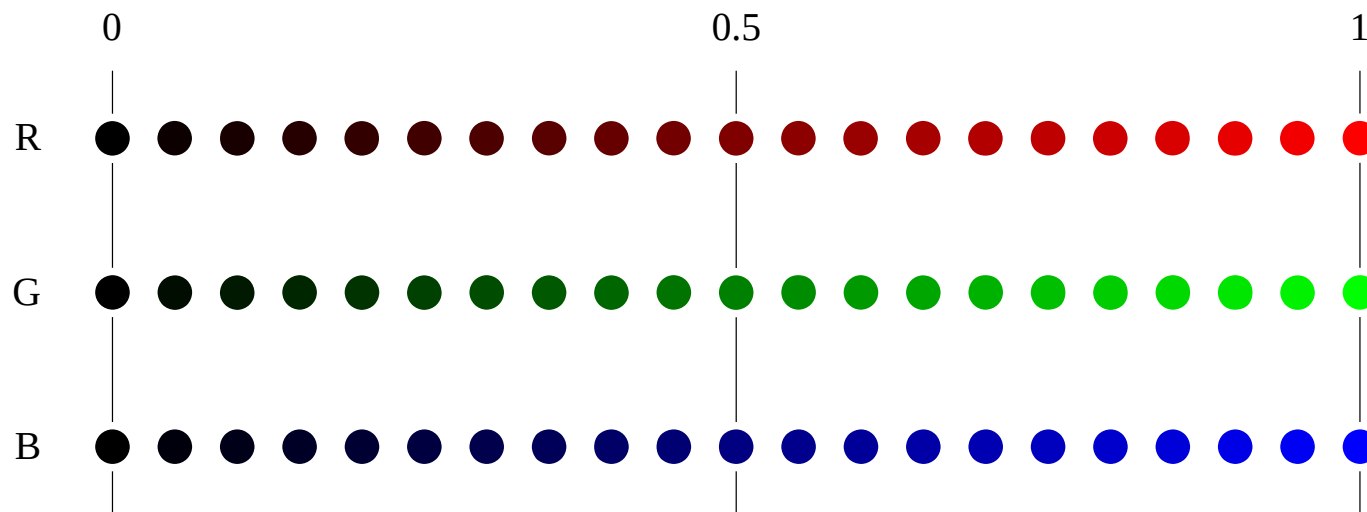
Same Length, Different Direction

- Vectors with equal magnitude but different directions represent different physical quantities.
- Vectors at right angles are called perpendicular, or orthogonal — each is independent of the other.
- Vectors pointing in opposite directions can cancel each other out.
- Placing equal-length vectors tail-to-tail, their tips all land on the same circle.



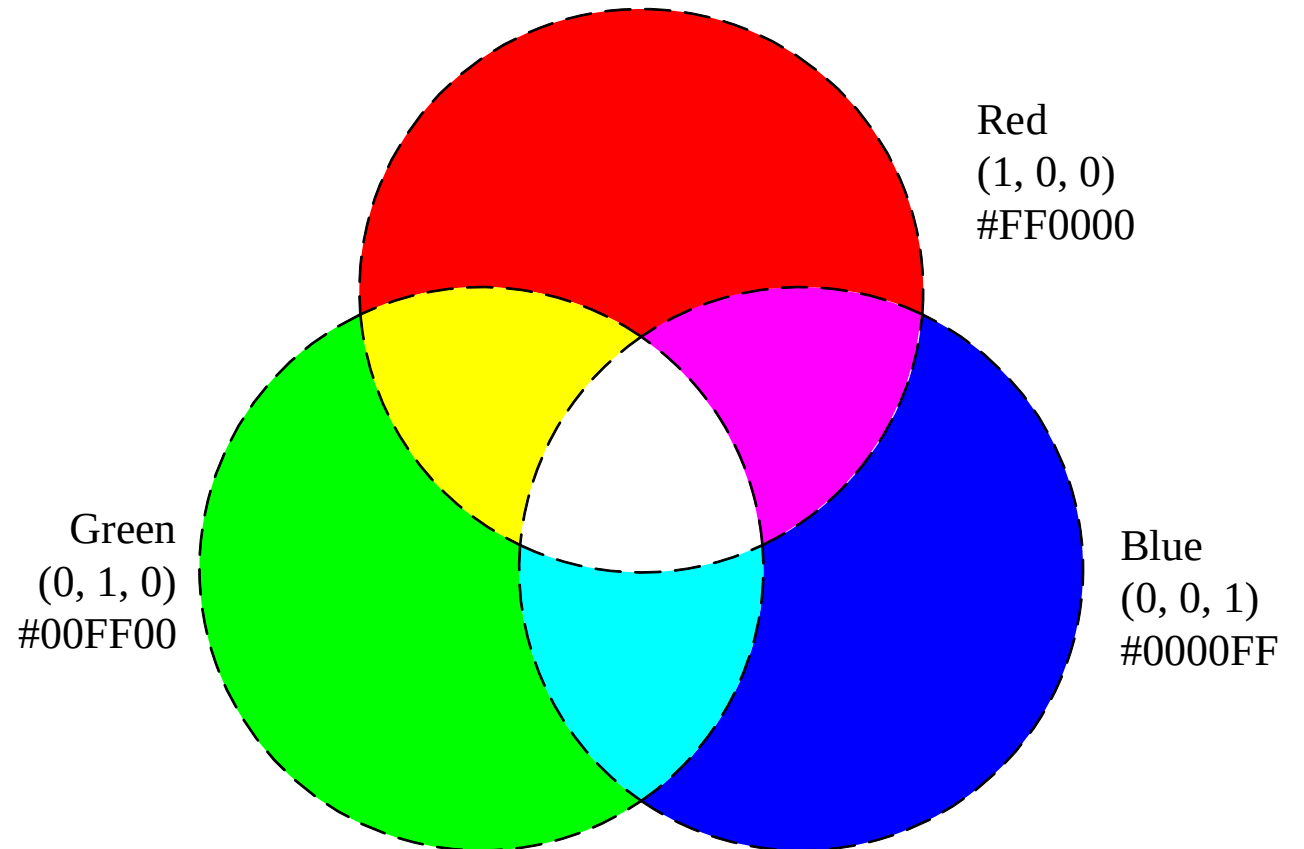
Color Is a Vector, Too

- A color in the RGB model is a 3D vector with three components: R (red), G (green), B (blue).
- Vector dimension: the number of independent components needed to describe it — a 2D vector has 2, a 3D vector has 3.
- In this book each component ranges over $[0, 1]$; in practice, integers over $[0, 255]$ are also common.



Additive Mixing: Light Adds Up

- Red + Green = Yellow light; Blue + Green = Cyan; Red + Blue = Magenta.
- Red, Green, and Blue at full intensity together produce White — every visible color superimposed.
- Note: RGB mixes light, not pigment.
- Extension: this additive principle powers displays, TVs, and projectors.



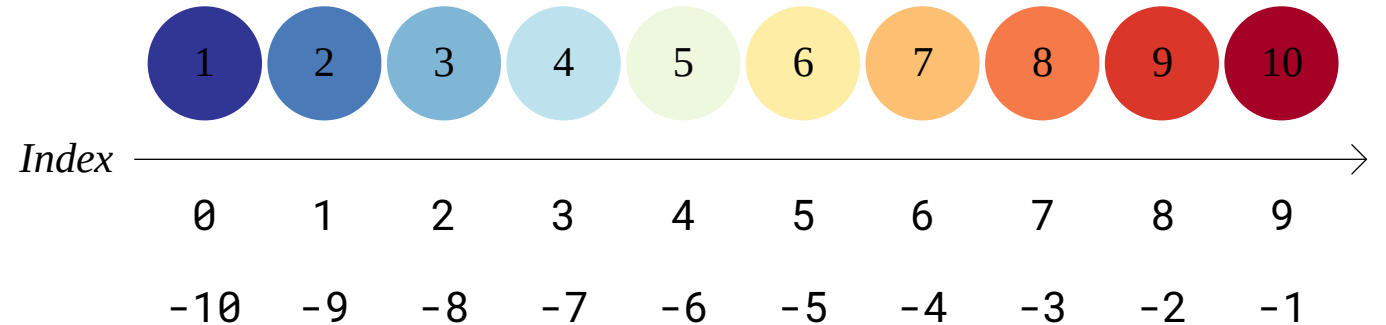
Components Laid Out in a Row

$$\mathbf{a} = [a_1 \ a_2 \ \cdots \ a_n]$$

- A row vector arranges its components (entries) along a single row.
- The component at index i is written a_i , where i runs from 1 to n .
- Vector dimension is the number of components — the vector above is n -dimensional.
- In machine learning, one data sample is often represented as a row vector.

Zero-Indexing vs. Math Indexing

- NumPy arrays are zero-indexed: counting starts at 0, so the first element sits at index 0.
- Mathematics conventionally starts at 1: a vector's first component has index 1.
- Negative indices count from the end: -1 is the last component, -2 the second-to-last, and so on.



Creating, Indexing, and Slicing a 1D Array

Code 1. 1D Array | LA_01_01_01.ipynb

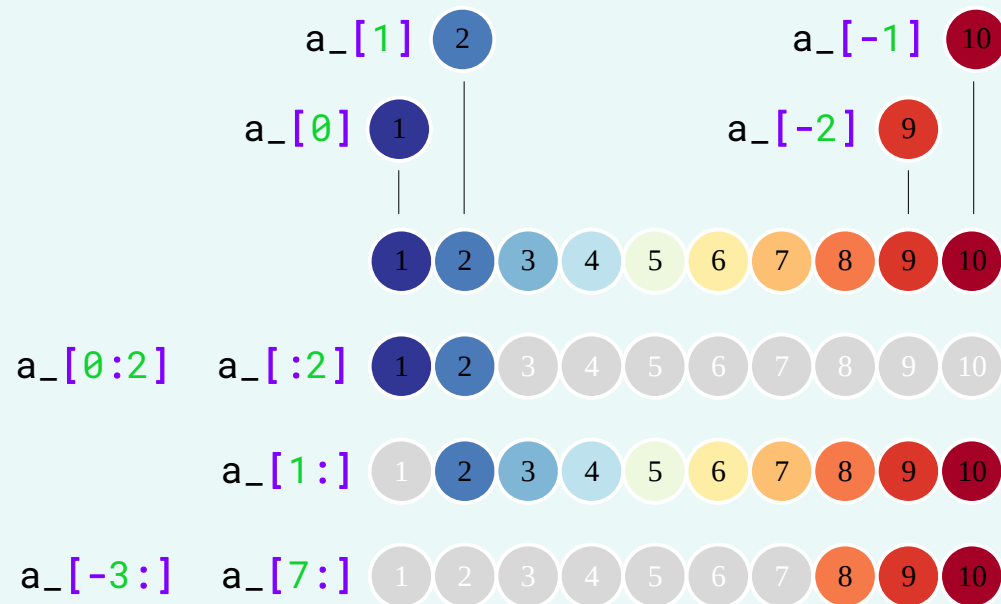
- `numpy.array()` converts a Python list into a NumPy array.
- `.ndim` returns the number of axes; `.shape` returns the size along each axis.
- Indexing `a_[i]` returns one element; slicing `a_[start:end]` excludes the end index.
- Negative indices/slices count from the end of the array.

初始化
a `import numpy as np`

创建数组
b `a_list = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]`
c `a_ = np.array(a_list)`
d `# a_ = np.arange(1,11)`
e `dim = a_.ndim`
f `shape = a_.shape`

索引
g `a_[0]`
h `a_[1]`
i `a_[-1]`
j `a_[-2]`

切片
k `a_[0:2]`
l `a_[1:2]`
m `a_[1:]`
n `a_[-3:]`
o `a_[7:]`

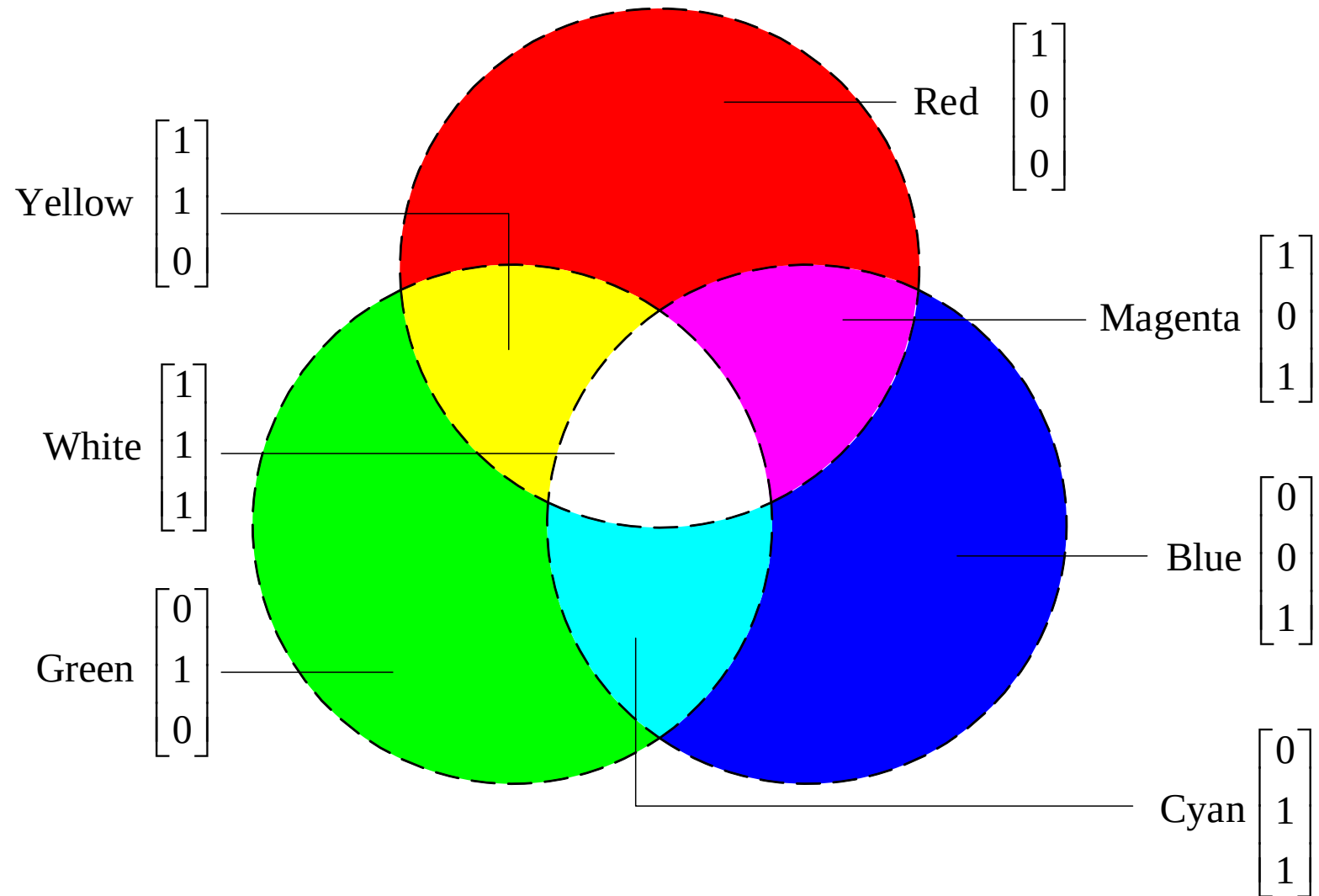


Components Laid Out in a Column

$$\mathbf{a} = [3, 4]^T \text{ (same as } [3; 4] \text{)}$$

- A column vector arranges its components along a single column; unless stated otherwise, vectors in this book default to column vectors.
- Column vectors are written in lowercase, bold, italic letters: $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$, $\mathbf{u}$, $\mathbf{v}$, $\mathbf{x}$, $\mathbf{y}$.
- To save vertical space, this book often writes a column vector via transpose, e.g. $\mathbf{a} = [3, 4]^T$, also written $[3; 4]$.

Seven Colors, Seven Column Vectors



Two Special Vectors + One Key Insight

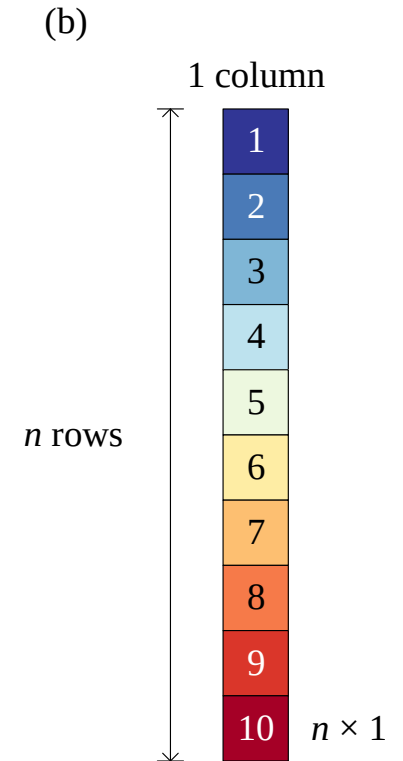
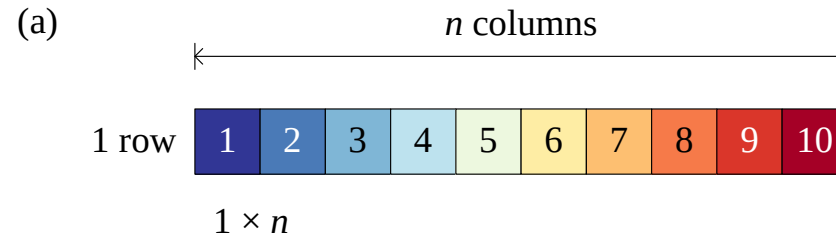
All-ones vector 1: every component equals 1 — this is white, $(1, 1, 1)$.

Zero vector 0: every component equals 0 — this is black, $(0, 0, 0)$.

Vector addition: Red+Blue=Magenta, Blue+Green=Cyan, Red+Green=Yellow, Red+Green+Blue=White — these color combinations are, at heart, vector addition.

Row Vector = $1 \times n$ Matrix, Column Vector = $n \times 1$ Matrix

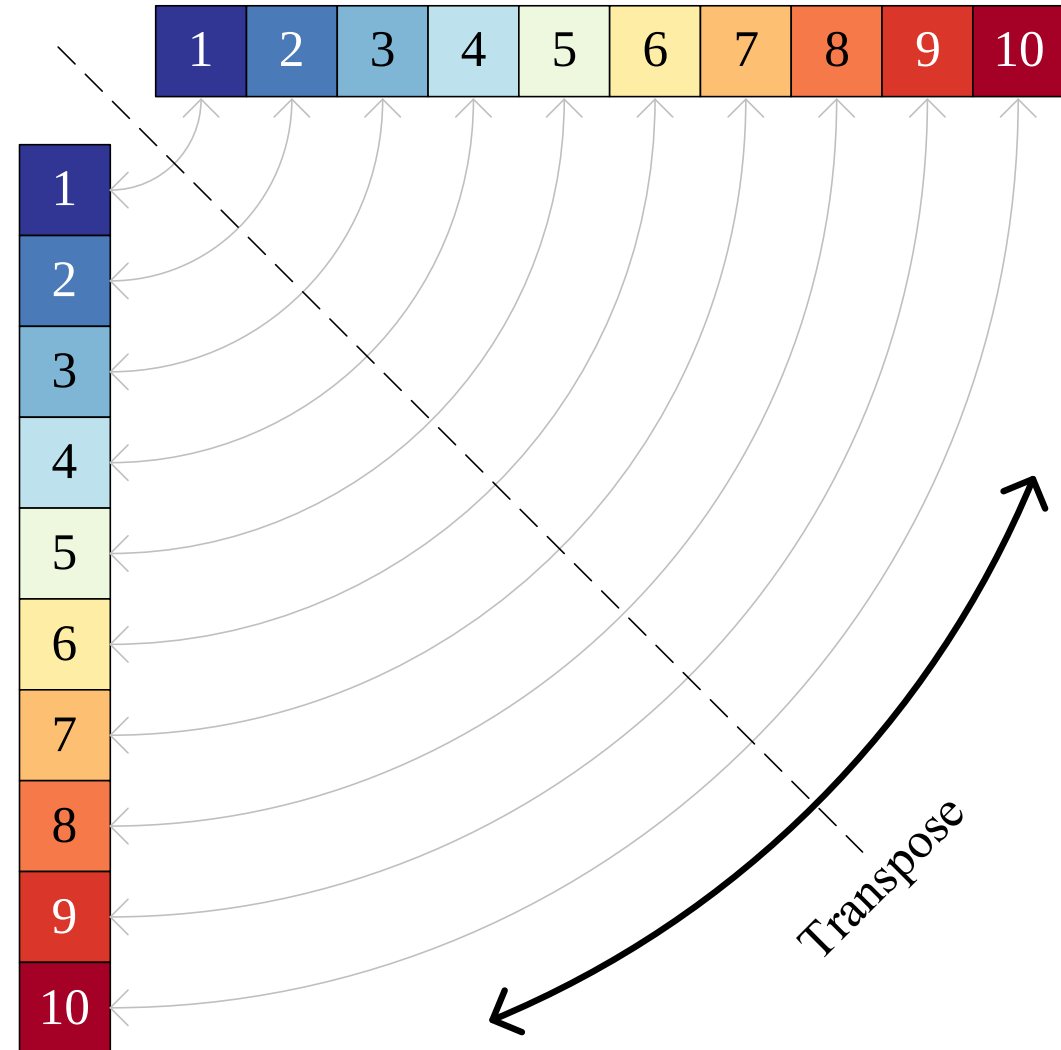
- A matrix is a rectangular array of rows and columns; each entry is fixed by a row index and a column index — row index always comes first.
- A row vector $\mathbf{a}^{(1)}$ is a matrix with one row and many columns, shape $1 \times n$.
- A column vector $\mathbf{a}_1$ is a matrix with many rows and one column, shape $n \times 1$.



TRANSPOSE

Swapping Rows and Columns: a^T

- Transpose swaps a matrix's rows and columns, denoted with a superscript T.
- A transposed row vector becomes a column vector, and a transposed column vector becomes a row vector.
- Transpose changes neither the number of components nor their relative order.



Using 2D Arrays to Represent Vectors

Code 2. 2D Array | LA_01_01_02.ipynb

- `np.array([[1,...,10]])` builds a 1×10 row vector (a 2D array, `ndim = 2`).
- `row_vector.T` transposes it into a 10×1 column vector.
- 2D indexing uses `[row, col]`; slicing also supports negative indices.

```
## 初始化
import numpy as np

## 创建行向量 (二维)
a row_vector = np.array([[1, 2, 3, 4, 5, 6, 7, 8, 9, 10]])
row_vector.ndim
row_vector.shape

### 索引
b row_vector[0, 0]
row_vector[0, 1]
row_vector[0, -1]

### 切片
c row_vector[:, :2]
row_vector[:, 1:]
row_vector[:, -3:]

## 创建列向量 (二维)

### 转置
d col_vector = row_vector.T

### 索引
e col_vector[0, 0]
col_vector[1, 0]
col_vector[-1, 0]

### 切片
f col_vector[:, 2, :]
col_vector[1:, :]
col_vector[-3:, :]
```

Key Terms to Remember

Vector 向量

Vector field 向量场

Direction 方向

Column vector 列向量

Matrix 矩阵

Scalar 标量

Magnitude / length 大小 / 长度

Row vector 行向量

Transpose 转置

Dimension 维数

KEEP EXPLORING

Companion Resources & Further Reading

Open-Source Code & Manuscript

All code, Jupyter notebooks, and PDF drafts: <https://github.com/Visualize-ML>

“Python for Beginners” (Book1_Python-For-Beginners)

For Python, Anaconda, and JupyterLab basics, see this companion volume

“The Beauty of Visualization”

For visualizing math and data, see this companion volume in the series

Companion Video Lectures

Bilibili — DrGinger: space.bilibili.com/513194466

Complete These with the Help of DeepSeek / ChatGPT

Q1 Install Anaconda and learn JupyterLab (see “Python for Beginners,” ch. 1-3).

Q3 From array `[1,2,3,4,5,6]`, extract the odd and the even entries.

Q5 Transpose 1D array `[1,...,6]` and explain the result.

Q7 Learn `linspace()`, `zeros()`, `ones()`, and `random.rand()`.

Q9 Have AI annotate a streamplot example line by line; learn to draw one.

Q2 Study Python's list, tuple, and variable concepts.

Q4 From 2D array `[[-3,...,3]]`, extract entries >0 , ≥ 0 , <2 , and ≤ 2 .

Q6 Transpose 2D array `[[-3,...,3]]` and explain the result.

Q8 Learn how to define colors in Matplotlib.

Q10 Have AI annotate a quiver3d example line by line; learn to draw one.

Hands-On Vector Practice with Python

Q11 Review Python's list, tuple, and variable concepts.

Q13 Build an evenly-spaced vector with `numpy.arange(0,20,2)`; reshape it into a 5×2 array.

Q15 Create three vectors with `zeros(5)`, `ones(5)`, and `random.rand(5)`; print each.

Q17 Generate two random RGB column vectors, add them in code, and check the sum lies in `[0,1]`.

Q19 Use Matplotlib to draw 2D vector (3,4) as an arrow (hint: `plt.quiver()`).

Q12 Create `a=[2,-1,5,3,0,7]` with `numpy.array()`, then print its `ndim` and `shape`.

Q14 Write code to extract all components divisible by 3 from `[10,20,...,100]`.

Q16 Write a function `magnitude(v)` that computes any vector's length (hint: `numpy.linalg.norm`).

Q18 Define row vector `a=numpy.array([[1,2,3,4,5]])` and verify `a.T.T` equals `a`.

Q20 Extract the components at even indices (0,2,4,...) from `numpy.arange(1,21)`.

Hands-On Vector Practice with Python

Q21 Create a 3×1 column vector, then flatten it with `.flatten()` or `.reshape(-1)`.

Q23 Generate a vector of 100 random numbers; count how many exceed 0.5.

Q25 Build a dict mapping color names to RGB vectors (e.g. "red": [1,0,0]); print each one.

Q27 Use `numpy.linspace(0,1,11)` to build an 11-step gradient for RGB's R component.

Q29 Generate a random 6D row vector; slice it into two new vectors — first 3 and last 3.

Q31 Capstone: write `describe_vector(v)` to print a vector's dimension, shape, magnitude, and zero-check.

Q22 Write code to test whether $u=[1,0]$ and $v=[0,1]$ are orthogonal (hint: `numpy.dot(u,v)`).

Q24 Convert row vector [5,10,15,20] into a column vector; print the shape before and after.

Q26 Write code verifying that adding the zero vector leaves any vector unchanged.

Q28 From `[[1,2,3],[4,5,6]]`, extract row 1 and column 2 separately; state each result's ndim.

Q30 Compare the ndim and shape of `numpy.array([1,2,3])` vs. `numpy.array([[1,2,3]])`.

10 Prompts to Explore with AI

1

Using three new real-life examples I haven't seen (not wind, water flow, or displacement), explain the difference between a scalar and a vector.

2

Given the coordinates of a 2D vector's initial and terminal points, calculate its magnitude and direction angle, showing every step.

3

Explain why wind and water flow are described by vector fields while temperature is a scalar field — give one new example of each.

4

Write Python/NumPy code that builds an RGB color as a column vector and renders the corresponding color swatch.

5

Explain why “orthogonal” and “perpendicular” mean the same thing for 2D vectors — do they still coincide in higher dimensions?

10 Prompts to Explore with AI

- 6 Design a short exercise where I use NumPy to create a 1×5 row vector and a 5×1 column vector, then verify the transpose relationship.

- 7 From the matrix viewpoint, explain why a row vector has shape $1 \times n$ and a column vector has shape $n \times 1$ — describe it clearly.

- 8 Give a concrete machine-learning example where a data sample is a row vector, and explain what each component means.

- 9 Explain NumPy's zero-indexing and negative-indexing rules, then quiz me with five indexing/slicing practice problems.

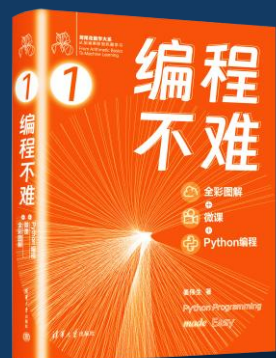
- 10 Turn this section's core concepts (magnitude, direction, dimension, row/column vectors, transpose) into a text outline for a mind map.

数学

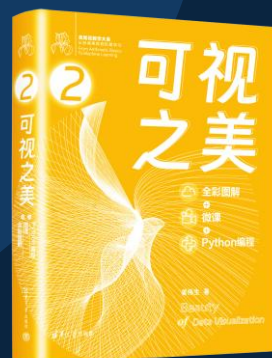
数学基础

线性代数

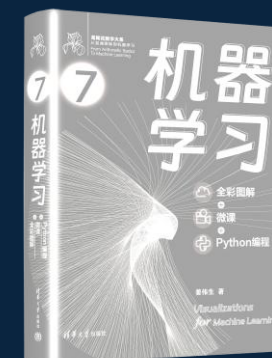
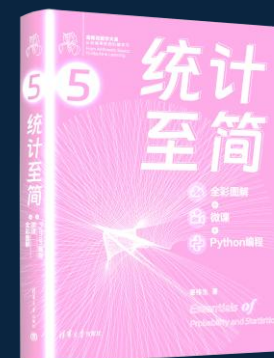
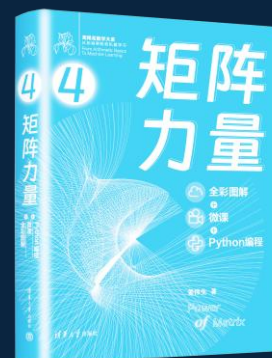
概率统计



Python编程



数据可视化



工具

数据分析
图和网络

回归、降维
分类、聚类

实践

